higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## BUILDING AND STRUCTURAL SURVEYING N6

## 21 NOVEMBER 2019

**This marking guideline consists of 9 pages.**

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## QUESTION 1

1.1 Back sight is the first sight taken after setting up the levelling instrument anywhere.

1.2 A field book is a small book used on site to enter or record data during execution of survey work.

1.3 Chaining is the common site name given to all taping work.

1.4 Chainage is the position of a point from the origin.

1.5 Isogonic lines are imaginary lines joining places on the earth's surface where the magnetic declination is the same at a given time.

(5 × 2) **[10]**

**QUESTION 2**

| NAME | JOIN | ΔY | ΔX | NAME | Y | X |
|---|---|---|---|---|---|---|
| A | | | | A | +4 049,94 | +6 803,68 |
| 209:47:34 | | -68,46 | -119,57 | | | |
| 137,78 m | | -0,06 | -0,04 | | | |
| | | | | | | |
| C1 | | | | C1 | +3 981,42 | +6 684,07 |
| 126:41:26 | | +140,62 | -104,78 | | | |
| 174,36 m | | -0,08 | -0,05 | | | |
| | | | | | | |
| C2 | | | | C2 | +4 121,96 | +6 579,24 |
| 99:19:36 | | +276,75 | -45,45 | | | |
| 280,46 m | | -0,13 | -0,09 | | | |
| | | | | | | |
| C3 | | | | C3 | +4 398,58 | +6 533,70 |
| 214:10:38 | | -102,07 | -150,32 | | | |
| 181,70 m | | -0,09 | 0,06 | | | |
| | | | | | | |
| E | | | | E | +4 296,42 | +6 383,32 |
| 1407,52✓ | | +246,84√ | -420,12√ | | +246,48√ | -420,36√ |
| | | +246,48√ | -420,36√ | | | |
| | | -0,36√ | -0,24√ | | | |

$\frac{-0,36}{775,30}$ X leg√ $\frac{-0,24}{775,30}$ X leg√

1 × 7 = 7
½ × 26 = 13 **[20]**

## QUESTION 3

| Station | | Distance | | HI or middle hair MH | Angles | | HI – MH + - | Height component + - | Height difference + - | Elevation of point | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|
| From | To | Stadia | Hor | | Hor | Vert | | | | | |
| A | | | | 1,43 | | | | | | 231,99✓ | |
| | A1 | 2,98<br>1,28 | 167,59√ | 2,13✓ | 14:38:00 | 83:10:00 | -0,7✓ | +20,08√ | +19,38✓ | 251,37✓ | |
| | A2 | 3,84<br>1,20 | 263,55√ | 2,52✓ | 96:20:00 | 92:22:00 | -1,09✓ | -10,89√ | -11,98✓ | 220,01✓ | |
| | A3 | 3,62<br>1,96 | 161,41√ | 2,79✓ | 206:52:00 | 99:34:00 | -1,36✓ | -27,20√ | -28,56✓ | 203,43✓ | |
| | A4 | 2,62<br>1,02 | 156,72√ | 1,82✓ | 258:32:00 | 81:46:00 | -0,39✓ | +22,68√ | +22,29✓ | 254,28 | BM 1 |

1 × 16 = 16
½ × 8 = 4 **[20]**

## QUESTION 4

Distance R1 – R2 = 3 122,24 – 2 932,82
= 189,42 m

Area R1:

$$W_1 = \frac{g\ (a + hs)}{g - s}$$
$$= \frac{6\ (4{,}5 + (2{,}68\ \times\ 2{,}5))}{6\ -\ 2{,}5}\checkmark$$
$$= 19{,}2\checkmark$$

$$W_2 = \frac{g\ (a + hs)}{g - s}$$
$$= \frac{67{,}2}{8{,}5}\checkmark$$
$$= 7{,}9\checkmark$$

$$\text{Area} = \frac{(w1\ \times\ w2) - a2}{s}$$
$$= \frac{(19{,}2\ \times\ 7{,}9) - (4{,}5)2}{2{,}5}\checkmark$$
$$= 52{,}57\ m^2\checkmark$$

Area R2:

$$W_1 = \frac{g\ (a + hs)}{g - s}$$
$$= \frac{8\ (4{,}5 + (1{,}76\ \times\ 2{,}5))}{8\ -\ 2{,}5}\checkmark$$
$$= 12{,}95\checkmark$$

$$W_2 = \frac{g\ (a + hs)}{g - s}$$
$$= \frac{71{,}2}{10{,}5}\checkmark$$
$$= 6{,}78\checkmark$$

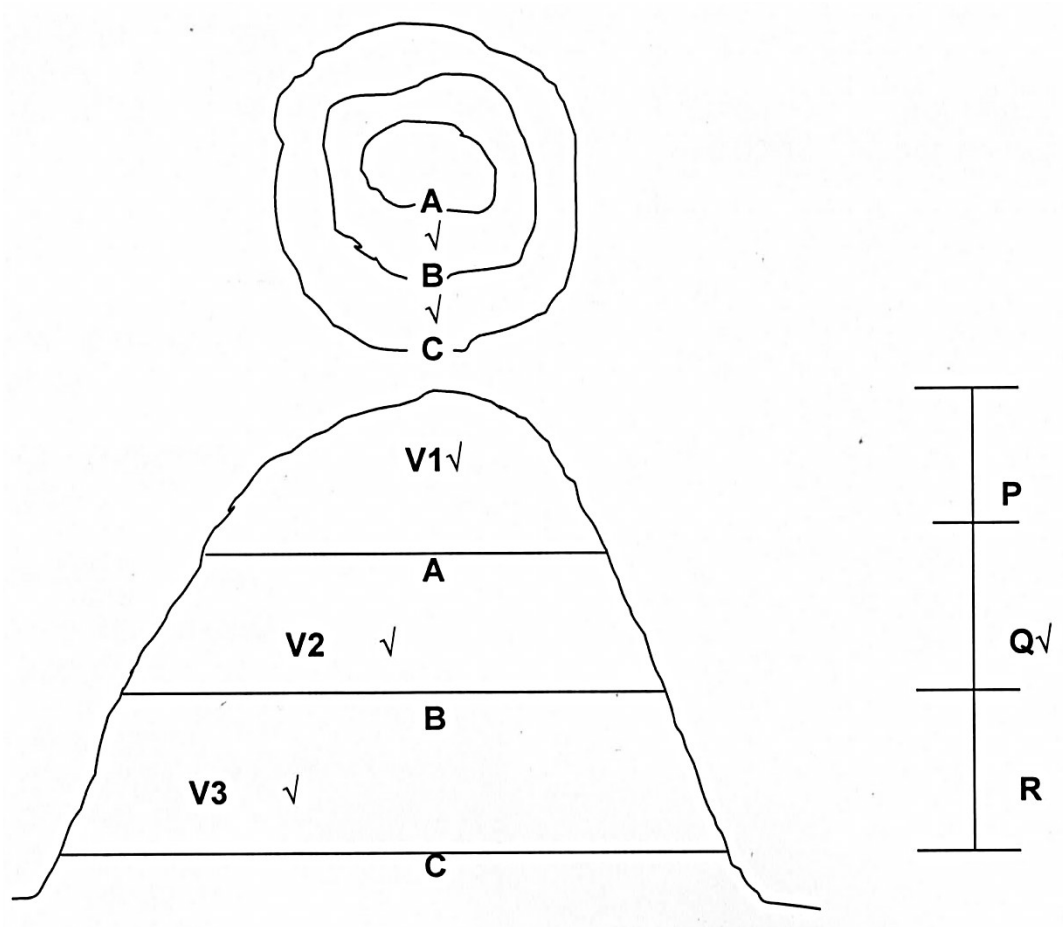
$$\text{Area} = \frac{(W_1 \times W_2) - a^2}{S}$$
$$= \frac{(12{,}95 \times 6{,}78) - (4{,}5)^2}{2{,}5}\checkmark$$
$$= 27{,}02\ m^2\checkmark$$

$$\text{Volume} = \frac{A_1 + A_2}{2} \times \text{distance}$$
$$= (\underline{52{,}57 + 27{,}02}) \times 108{,}42\checkmark$$

**[14]**

$$= 7\,537{,}96\ \text{m}^3 \checkmark$$

**QUESTION 5**




A, B and C are the contour lines.√
P, Q and R are the vertical intervals.√
Measure the areas of A, B and C with a planimeter.√

½ A × P = volume V1√

$^{A+B}/_{2}$ × Q = volume V2√

$^{B+C}/_{2}$ × R = volume V3√ (½ × 12) **[6]**

**QUESTION 6**

At S1:

| Station | Circle left | Circle right | Mean angle | Correction | Adjusted angle |
|---|---|---|---|---|---|
| | | | | | |
| A | 256:46:28 | 76:45:48 | | | |
| B | 310:56:14 | 130:56:24 | | | |
| | 54:09:46 | 54:10:12 | 54:09:59 | +00:00:06 | 54:10:05✓ |
| | | | | | |
| B | 310:56:14 | 130:56:24 | | | |
| C | 88:28:10 | 268:27:34 | | | |
| | 137:31:56 | 137:31:34 | 137:31:45 | +00:00:06 | 137:31:51✓ |
| | | | | | |
| C | 88:28:10 | 264:27:34 | | | |
| A | 256:46:12 | 76:45:28 | | | |
| | 168:18:02 | 168:17:54 | 167:17:58 | +00:00:06 | 168:18:04✓ |
| | | | 359:59:42 | +00:00:18 | 360:00:00✓ |
| | | | | | |

Join S1–B:

$Dy_{S1-B}$ = -1 923,26 – (-1 467,36)
= -455,9✓

$Dy_{S1-B}$ = +2 012,43 – (+2 931,21)
= -981,78✓

Direction S1–B = $\frac{-455,9}{-918,78}$ 3rd quadrant
Tan$\theta$ = 0,496201484✓
$\theta$ = 206:23:26

Direction S1–B = 206:23:26
Angle B;S1;C = 137:31:51✓
S1–C = 343:55:17
Angle C;S1;A = 168:18:04✓
S1–A = 152:13:21
Angle A;S1;B = 54:10:05
S1–B = 206:23:26✓ **[10]**

**QUESTION 7**

$T = R \tan \frac{\Delta}{2}$ √

$T = 210 \times \tan 21:00:00$ √

$T = 80{,}61$ m√

BC = 4 348,21 – 80,61√
= 4 267,6 m√

$La = \frac{\pi \Delta R}{180}$

$= \frac{\pi \times 42:00:00 \times 210}{180}$ √

= 153,93 m√

EC = BC + La

= 4 267,6 + 153,93√

= 4 421,53 m√

$(a = 12{,}4) = \frac{1\,718{,}9 \times 12{,}4}{210 \times 60} = 01:41:29{,}82$ √

$(a = 20{,}00) = \frac{1\,718{,}9 \times 20{,}00}{210 \times 60} = 02:43:42$ √

$(a = 1{,}53) = \frac{1\,718{,}9 \times 1{,}53}{210 \times 60} = 00:12:31$ √

| CHAINAGE | CHORD | $\vartheta$ | $\alpha$ |
|---|---|---|---|
| BC 4 267,60 | | 00:00:00 | 360:00:00✓ |
| 4 280,00 | 12,40 | 01:41:30 | 358:18:30✓ |
| 4 300,00 | 20,00 | 02:43:42 | 355:34:48✓ |
| 4 320,00 | 20,00 | 02:43:42 | 352:51:06✓ |
| 4 340,00 | 20,00 | 02:43:42 | 350:07:24✓ |
| 4 360,00 | 20,00 | 02:43:42 | 347:23:42✓ |
| 4 380,00 | 20,00 | 02:43:42 | 344:40:00✓ |
| 4 400,00 | 20,00 | 02:43:42 | 341:56:18✓ |
| 4 420,00 | 20,00 | 02:43:42 | 339:12:36✓ |
| EC 4 421,53 | 1,53 | 00:12:31 | 339:00:05✓ |
| | | | |
| 153,93✓ | 153,93✓ | 20:59:55✓ | 20:59:55✓ |

½ × 12 = 6
1 × 14 = 14 **[20]**

**TOTAL: 100**

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